

Title

Data and scripts associated with “Moisture content modulates DOM thermodynamic regulation of oxygen consumption in drying streambed sediments”

Summary

This data package is associated with the publication “Moisture content modulates DOM thermodynamic regulation of oxygen consumption in drying streambed sediments” published in Scientific Reports (Garayburu-Caruso et al., 2026). The package contains processed data products and scripts used to quantify how drying and re-inundation of riverbed sediments influence dissolved organic matter (DOM) thermodynamic properties and their relationship with sediment oxygen (O_2) consumption across 33 stream sites in the contiguous United States. The data package contains DOM thermodynamic metrics (e.g., Gibbs free energy of carbon oxidation and thermodynamic efficiency), and O_2 consumption along with watershed-scale climate and land-cover metrics used as explanatory variables in the analyses.

Underlying unprocessed and processed ultrahigh-resolution mass spectrometry data, oxygen consumption rates from laboratory moisture-manipulation experiments, within-sample environmental properties, sediment moisture content and contextual field measurements are archived separately at <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/2428003> (Laan et al., 2024) and <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/1923689> (Forbes et al., 2023).

A preliminary version of this data package was published in February 2-26 at the time of manuscript submission. It was updated in June 2026, at the time of manuscript acceptance, to include the finalized data and additional metadata (this readme, data dictionary, and file level metadata).

For details on how to navigate data packages generated by this project, see <https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA/About>.

Brief Overview of Methods

For a full description of the methods, see associated manuscript. Briefly, shallow riverbed sediments (1–3 cm) and filtered river water were collected at WHONDRS sites across the contiguous United States between April 2022 and October 2023 and shipped to Pacific Northwest National Laboratory. In the laboratory, homogenized sediments from each site were either allowed to dry (dry treatment) or kept inundated (wet treatment) for 21-days, followed by re-inundation with native water to quantify sediment oxygen (O_2) consumption rates using optical oxygen sensors. Dissolved organic matter (DOM) chemistry of post-incubation water was analyzed using solid-phase extraction and ultrahigh-resolution Fourier transform ion cyclotron resonance mass spectrometry (FTICR-MS). Molecular formulas were processed with CoreMS and ftmsRanalysis to calculate DOM thermodynamic metrics (Gibbs free energy of carbon oxidation, ΔG_{cox} , and thermodynamic efficiency, λ) using unique molecular formulas within the wet and dry treatments at each site. Sediment physical and geochemical properties and watershed-scale climatic and land-cover variables were combined with DOM thermodynamics and oxygen consumption data and used in clustering, generalized additive models, and breakpoint analyses to evaluate how moisture modulates thermodynamic regulation of sediment respiration.

Data Package Structure

In addition to this readme, this data package also includes a file-level metadata (FLMD) file that describes each file and a data dictionary (DD) that describes all column/row headers and variable definitions.

At the top level, the data package is organized into five main folders: (1) Data, (2) Figures, (3) Map, (4) GAM_Reulsts, and (5) src. The Data folder contains analysis-ready tabular files with oxygen consumption rates, DOM thermodynamic properties by site and treatment, site-level environmental variables, watershed-scale metrics, and other derived variables referenced in the manuscript. The Figures folder contains static image files associated with the main text and supplemental figures, while the Map folder includes spatial data and map-layer files used to create the sampling-location map. The GAM results folder contains the results for each of the general additive model (GAM). The src folder contains R scripts used to perform data processing, statistical analyses (including clustering, generalized additive models, and threshold analysis), and figure generation. This data package is associated with a GitHub repository found at https://github.com/WHONDRS-Hub/ECA_DOM_Thermodynamics.

Citations, Acknowledgements, and License

Acknowledgements

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Cite this data package with the appropriate DOI. Cite the associated manuscript in any work that uses analyses or conclusions presented in the manuscript

Citations

- Laan, M. et al. (2024). Laboratory time series moisture manipulative experiment from sediment across the contiguous US: time series aerobic respiration and geochemistry. ESS-DIVE repository. <https://data.ess-dive.lbl.gov/view/doi:10.15485/2428003>
- Forbes, B. et al. (2023). WHONDRS River Corridor Dissolved Oxygen, Temperature, Sediment Aerobic Respiration, Grain Size, and Water Chemistry from Machine-Learning-Informed Sites across the Contiguous United States (v4). ESS-DIVE repository. <https://data.ess-dive.lbl.gov/view/doi:10.15485/1923689>
- Garayburu-Caruso, V.A., Danczak, R.E., Stegen, J.C., Renteria, L., McCall, M., Goldman, A.E., Chu, R.K., Toyoda, J., Resch, C.T., Torgeson, J.M., et al. (2020). Using community science to reveal the global chemogeography of river metabolomes. *Metabolites*, 10(12), 518. <https://doi.org/10.3390/metabo10120518>

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Change History

Change history:

Data Package Version	Changes
Preliminary <i>February 2026</i>	Preliminary data package publication
Version 1 <i>June 2026</i>	<u>NEW FILES</u> • readme_Garayburu-Caruso_2026_ECA_Thermodynamics.pdf • Garayburu-Caruso_2026_ECA_Thermodynamics_dd.csv • Garayburu-Caruso_2026_ECA_Thermodynamics_flmd.csv <u>MODIFIED FILES</u> Data files were changed, added, and removed in response to manuscript reviewer comments. Briefly, we included data and scripts for the DOM thermodynamics and oxygen-consumption analyses, including site-level treatment summaries, watershed-scale and land-cover explanatory variables, generalized additive model outputs, moisture-threshold analyses, and figures. All changes were tracked in version control and can be reviewed in the GitHub commit history.